AQUAFAB INDUSTRIES

Running a Respirometry Trial

Intermittent-flow and closed respirometry, from setting up to saving the result.

For Aquafab Workbench 0.1.0

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Screenshots show simulated machines and modelled example data, not measurements. Aquafab Workbench is not a safety system.

aquafabindustries.com · sales@aquafabindustries.com · +1 (513) 781-8010

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1 Running a respirometry trial

This guide runs an intermittent-flow respirometry trial, the standard way to measure standard metabolic rate (SMR), from setting it up to saving the result. The same page also runs MMR, aerobic scope, P_{crit} and EPOC measurements; how they differ is at the end.

You need: a machine with a dissolved-oxygen probe, the probe calibrated (*Commissioning a New Machine*, step 5), the chamber's volume entered in **Settings → This machine's limits**, and a license to start a trial on connected hardware. Rehearse on a simulated machine first; it needs no license.

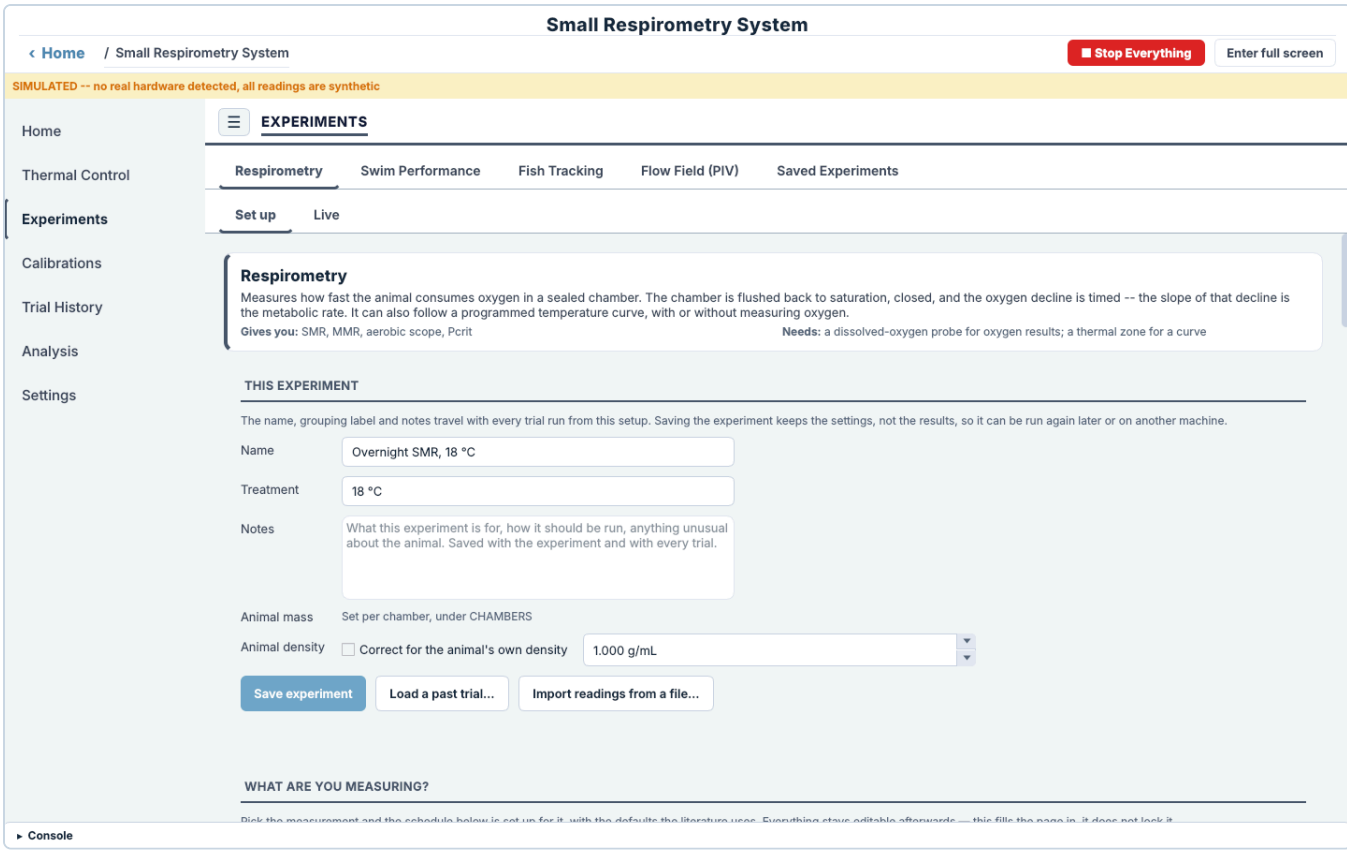
> Screenshots are from a simulated machine. Numbers in them are not measurements.

1.1 How intermittent-flow respirometry works

The chamber is **flushed** with fresh water back to saturation, left to **wait** a moment while the water mixes, then **closed** while the animal's oxygen uptake draws the oxygen down. The slope of that decline, times the water volume, divided by the animal's mass, is one MO_2 (mg O_2 kg^{-1} h^{-1}). Repeat for many cycles, and SMR is estimated from the low end of them (Chabot, Steffensen & Farrell 2016).

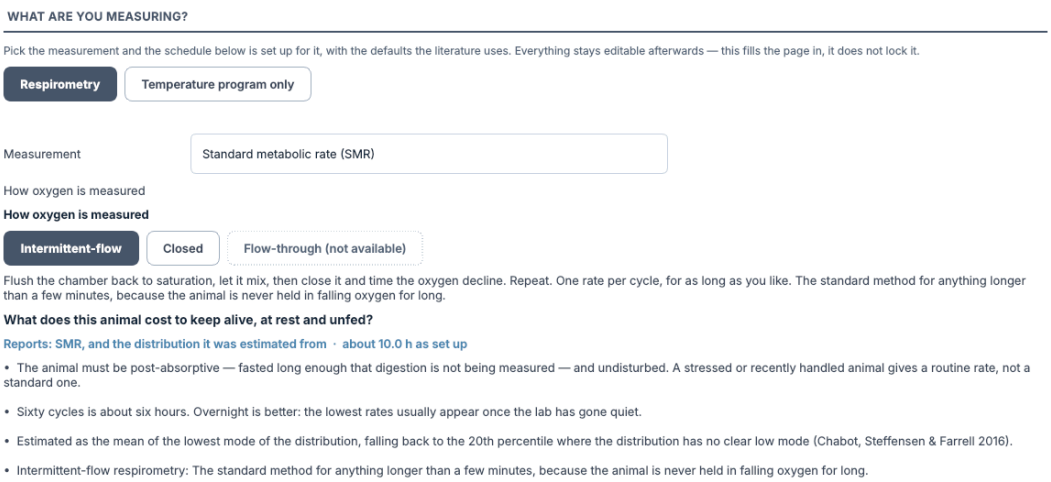
1.2 Set up

Open the machine, then **Experiments → Respirometry → Set up**. Work down the cards.



The top of the Set up view.

1.2.1 What are you measuring?

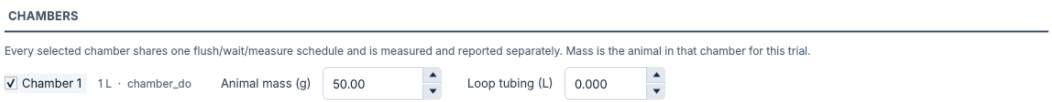


1. Choose **Respirometry** (not *Temperature program only*).
2. **Measurement: Standard metabolic rate (SMR)**. This fills in the literature defaults: 1 h acclimation, 60 cycles of 180 s flush, 60 s wait, 300 s measure, minimum R^2 0.95, SMR by MLND. About 10 hours. Everything stays editable.
3. **How oxygen is measured: Intermittent-flow.**

1.2.2 This experiment

- **Name**, for example *SMR overnight 18 °C*.
- **Treatment**, for example *18C*. Trials sharing a treatment are compared together in Analysis, so type it identically for every animal in a group.
- **Notes**: anything a reader would want to know.

1.2.3 Chambers



Tick each chamber in use and enter, for each:

- **Animal mass (g)**: this animal, weighed today.
- **Loop tubing (L)**: water in the chamber's recirculation loop, if any.

Read the line under the fields, for example *"0.953 L of water: 1.000 L chamber − 47 mL displaced by the animal"*. That is the volume every rate is multiplied by. If it looks wrong, the chamber volume in Settings is wrong.

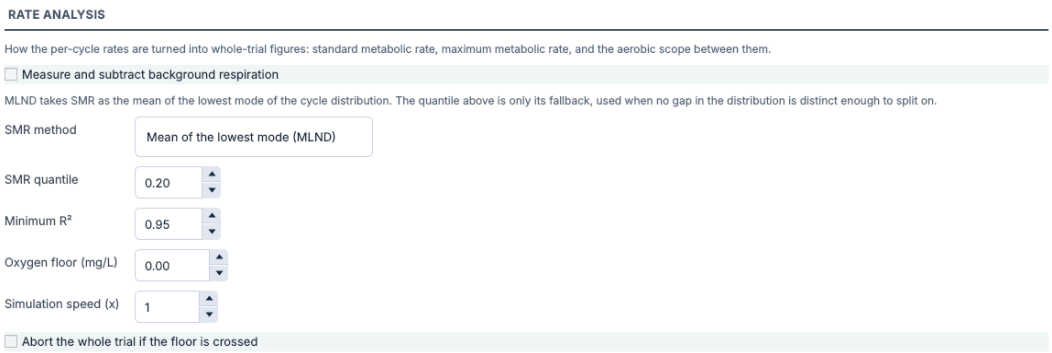
1.2.4 Acclimation, measurement cycle, water speed

- **Acclimation**: how long the animal settles before measuring, flushed to keep it oxygenated.
- **Measurement cycle**: **Cycles**, **Flush (s)**, **Wait (s)**, **Measure (s)**. Size the measure phase so oxygen falls usefully but not too far: a big animal in a small chamber needs a shorter measure. **Total protocol duration** adds it up.
- **Water speed**: **Leave as set**, or **Constant** to record the speed with the trial.

1.2.5 Temperature

Set **Constant temperature**. Leave **Use this curve…** unticked for a held-temperature SMR.

1.2.6 Rate analysis



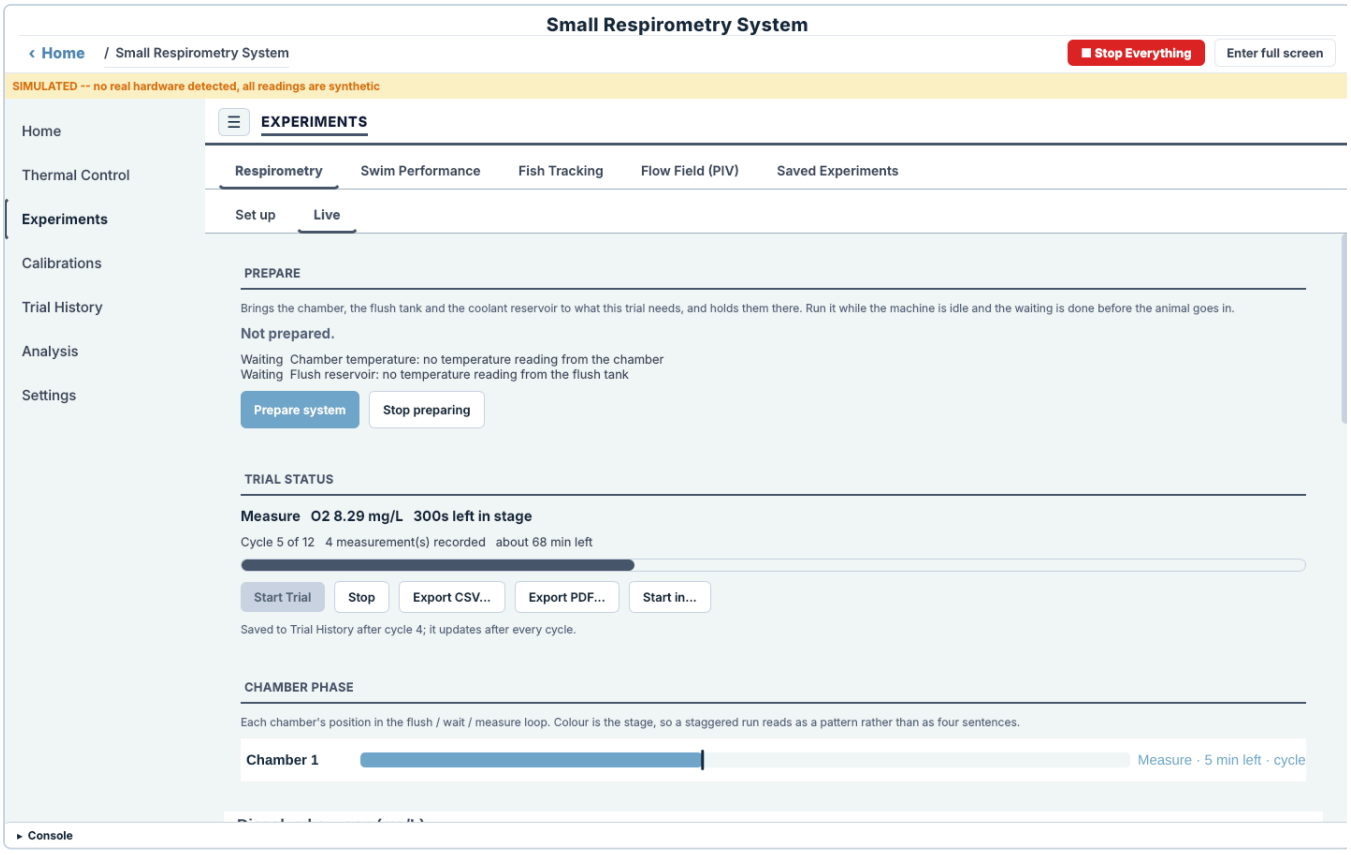
- **Measure and subtract background respiration**: adds a blank cycle and subtracts its rate (Svendsen et al. 2016). Recommended.
- **SMR method**: *Low quantile* or *MLND*; **SMR quantile** 0.2.
- **Minimum R^2**: cycles fitting worse are kept but excluded.
- **Oxygen floor (mg/L)**: if oxygen falls below this in a closed phase, the chamber is flushed at once. **Set it.**

Press **Save experiment** to keep the whole setup for the next animal.

1.3 Prepare the machine

Switch to **Live** and press **Prepare system**, before the animal goes in. The Prepare card lists what it is waiting for: the chamber at temperature, the flush tank at the same temperature (so a flush is not a thermal shock), the reservoir chilled. Wait until every line is ready.

1.4 Run



The Live view during a simulated trial.

1. Put the animal in, close the chamber, and check for bubbles.
2. Press **Start Trial** (**Ctrl+Return**). Or **Start in…** with a delay in minutes, to start after you have left the room.

3. Watch the first cycles. The **Dissolved oxygen** chart should fall steadily in each measure phase and recover with each flush; each cycle adds a point to **MO_2**. **Chamber phase** shows each chamber's stage.
4. **Derived statistics** update as cycles complete: SMR, MMR, aerobic scope, mean R^2.

If anything goes wrong, **Stop** ends the trial (completed cycles are kept), and ■ **Stop Everything** stops the whole machine.

1.5 Save

There is nothing to press: the trial saves itself to Trial History after the first cycle and again after every cycle, so a power cut at six in the morning keeps the night's cycles. The line under the buttons says how far it has got, and reads *"Trial saved to Trial History (N cycle(s))."* when the run ends. A run that never finished is marked *Interrupted* in Trial History. Optionally, **Export CSV…** (every cycle and the oxygen trace) or **Export PDF…** (a printed report).

The trial is now in **Trial History**, where **Inspect** shows every cycle and every setting used.

1.6 Other measurements on the same page

Measurement	What changes
Routine metabolic rate (RMR)	Shorter run; reports the mean of accepted cycles.
Maximum metabolic rate (MMR), after a chase	No acclimation, short cycles. Chase the animal to exhaustion, move it in, and start **immediately**.
Aerobic scope	SMR and MMR in one run, with absolute and factorial scope.
Critical oxygen level (P_{crit})	**Closed** respirometry: one long closed phase while oxygen falls. **Set the oxygen floor.** P_{crit} appears in Derived statistics once the oxygen range allows a fit.
Recovery after exercise (EPOC)	No acclimation, many short cycles following recovery.

1.7 Good practice

- Fast the animal as your protocol requires, and run overnight where you can.
- Check the respirometric volume line before every trial.
- In **Analysis → Work on a trial**, you can leave out the first cycles while the animal was settling, without changing the saved trial (*Analysing and Exporting Results*).